

Carry-Over Lottery Allocation: Practical Incentive-Compatible Drafts

Paper Track: Basketball

Open-source repository: <https://github.com/kvr06-ai/cola-sloan-ssac27>

Introduction

How can a league help weak teams through the draft without encouraging teams to tank? Carry-Over Lottery Allocation (COLA), presented here, is a family of draft mechanisms that all simultaneously discourage tanking and help the weakest teams.

Under any mechanism in the COLA framework, each team has a draft priority that rises and falls over time, but it does not rise based on more losses. That eliminates the tanking incentive. Draft priority carries over from year to year, being reduced when a team has playoff success or makes a high draft pick. That directs the most help to the weakest teams over time.

For 40 years, the NBA has navigated the tradeoff between helping weak teams and discouraging tanking. COLA solves that primary dilemma, but there are other tradeoffs worth considering among the COLA variants. After demonstrating COLA's incentive compatibility, this paper discusses the tradeoffs between mechanisms within the framework.

Methods

We identify several tunable dials for mechanisms within the COLA framework:

- Which teams are eligible for the lottery?
- How does a team's lottery priority increase?
- How does the priority decrease?
- How does the priority determine lottery odds and draft positions?

We consider different COLA variants plus other draft mechanisms, including 3-2-1 Lottery, and compare them using ZenGM Basketball-GM, an open-source simulator that plays out every game and roster move. We measure the mechanisms against these criteria:

- **Targeted help.** Does help reach the weakest teams, as measured by the difference between the help received by the highest priority team and other teams?
- **Long-term parity.** Can all teams be competitive, as measured by the longest drought any team has between playoff series wins?
- **Simplicity.** How complicated is the mechanism? Though measured subjectively, it is not difficult to identify complexity.
- **Robustness.** Does the mechanism admit tanking incentives at the margin? COLA variants are incentive-compatible under reasonable assumptions. Further relaxation of those assumptions may entail a choice between extra complexity or allowing potential tanking incentives in rare cases.

Results

Many mechanisms are dominated under these criteria, meaning there is another mechanism that is better on all four.

Using 30 simulations of 25 seasons for each of 9 mechanisms (6,750 seasons total), we identified a frontier of two viable mechanisms. Notably, one mechanism directly dominates both 3-2-1 Lottery and the 2019–2026 NBA Lottery on point estimates. The table summarizes the results.

Mechanism	Help to long-drought teams <i>draft places per drought season</i>	Reward for losing <i>draft places gained by finishing last</i>	Teams never winning a series <i>in 25 years, per league</i>	Longest drought <i>seasons, of 25</i>
Current NBA lottery (2019 to 2026) ¹	0.07	1.32	0.63	23.3
New NBA lottery, 3-2-1 (from 2027) ²	0.02	losing costs picks	1.20	24.4
Equal-odds lottery ³	0.00	0	0.93	23.7
Classic COLA, the original version ⁴	0.10	1.85	0.93	24.1
Waitlist COLA ⁵	0.13	0	1.10	24.4
Simple COLA ⁶	0.24	0	0.60	23.3
Full-lottery COLA ⁷	0.09	0	0.83	23.8
Countdown COLA ⁸	0.19	losing costs picks	1.33	24.6
Beckett COLA ⁹	0.28	losing costs picks	0.83	23.7

Table 1. Nine mechanisms on 30 simulated leagues of 25 seasons each, every mechanism on the same 30 starting leagues. *Help to long-drought teams:* among the fourteen teams that missed the playoffs, how many draft places earlier a team picks for each additional season since it last won a playoff series (a regression slope; the same clock for every mechanism). *Reward for losing:* take a season as played, drop the fifth-worst team to worst with everything else held, and measure how much its pick improves; zero for any rule that never reads record, and “losing costs picks” where losing pushes a team down the order. *Teams never winning a series:* of thirty teams, how many go all 25 seasons without a playoff series win, averaged over leagues. *Longest drought:* the longest stretch any one team goes without a series win. ¹The rule in force through the 2026 draft: the fourteen teams that miss the playoffs enter a lottery, four picks are drawn with worse records getting more chances, and picks 5 through 14 go in order of record, worst first. ²The rule adopted in May 2026 for the 2027 to 2029 drafts: sixteen teams, all sixteen picks drawn, the three worst records given fewer chances than the seven above them and never falling past pick 12. ³Every team that misses the playoffs has the same chance at every pick. ⁴Every team carries a waiting number that rises each season it misses the playoffs and falls when it succeeds; the top four picks are drawn with chances proportional to the waiting number, and picks 5 through 14 go by record. ⁵Classic COLA with one change: picks 5 through 14 go by waiting number, so no pick depends on record. ⁶No draw at all; the 22 teams that did not win a playoff series pick in order of waiting number. ⁷Every pick among the fourteen non-playoff teams is drawn, with chances proportional to the waiting number. ⁸Priority is the waiting number multiplied by wins, so a team out of the race gains by winning games; picks are assigned through a sequence of small elimination draws. ⁹Same priority as Countdown, open only to teams two or more seasons into a drought, with the top four picks drawn and the rest in priority order.

Conclusion

We identify a frontier of mechanisms that are not dominated and therefore viable options for implementation, depending on the league’s goals.